

Interview with Target Audience (Member of a family that plays a lot of board games): With the help of this interview, I learned about the games that they like to play, how often they like to play said games and the amount of space that they keep for board games and how many board games go there. Information like this helped me **design** the vague success criteria of the product.

Interview with Engineer: He advised on how to make the final product both functional and aesthetically pleasing. They may be able to provide insights into the best materials and components for the design, as well as advice on how to optimise the product for ease of use and durability.

Secondary Research:

After conducting the required research, I moved on to Secondary Research. For this section of the research, I mainly focused on the Aesthetics and Function.

Aesthetics of a product can be **measured** in a variety of ways. To begin, the product's overall design should be visually appealing and eye-catching. Using a variety of colours and designs can help the product stand out and capture the attention of the target audience. The finished product should be clean and polished, with no rough edges or visible flaws. Second, the design should be appropriate for the intended audience. For families with children over the age of six, the product should have a fun and playful design that appeals to children while also pleasing to adults. The use of bright colours and patterns can help make the product more appealing to children, whereas a more minimalist design with muted colours can help make the product more appealing to adults. Thirdly, the functionality of the product should be considered when evaluating its aesthetics. With clear instructions and an intuitive design, the product should be simple to **use** and navigate. The storage area should be well-organised and easily accessible, with game pieces and accessories arranged aesthetically and functionally.

Target Audience:

The target audience for this product are families with children over the age of six who enjoy playing board games. This product is intended to appeal to this demographic by offering them a convenient, interactive, and tangible way to store and play their favourite board games. Families with children over the age of six are likely to have a collection of board games that they enjoy playing with their children. However, because these board games take up a lot of space, they can be difficult to store. This product seeks to address this issue by combining several board games into a single compact design that takes up less space. This allows families to have access to a wide range of board games. Board games are a fun way for families to spend quality time together. This product is intended to improve on that experience by offering a visually appealing and interactive way to play multiple board games. Families can enjoy playing together without having to switch between multiple game boxes by combining different board games into one product.

Criterion B: Developing ideas

B1. **Develop** a list of design specifications/success criteria for the solution

Success Criterias provide clear guidelines and expectations for achieving a goal or objective. They **create** a shared vision of what success looks like and assist in measuring progress, tracking performance, and making adjustments. They assist in focusing efforts, staying on track, making data-driven decisions, and ensuring that resources are used effectively and efficiently to achieve desired results. And therefore, **creating** a success criteria for this solution was vital. Written below is the Success Criteria following the ACCESS FM format along with a **description** and its **justification**.

Criteria	Description	Justification/Connection with Research
Aesthetics	The board should include more than 4 board games. The board should include sufficient storage. This board should be a shade of brown to show the shade of the natural wood. Additionally, the board should include various colours so that it is not monotonous.	The purpose of this product is to combine multiple board games together. Additionally, this product caters to multiple age groups as it targets families. Children tend to be attracted to colour and hence it is essential that the product includes multiple colours. But it should not be too colourful that it comes off as

		too bright to an average person. And hence, there is a brown/beige border around the board game.
Cost	The cost of making the product should be less than \$16 approximately.	This cost has been derived from a website that mentions that a board game can cost \$7-\$25 per unit. The centre unit for the same has been used and this has been used to determine the cost of the product.
Customer	The customers for this product are families with 3 or more members. Anyone over the age of 3 years should be able to play this game. There will be small game pieces that will be harmful for children under the age of 3 and hence the age specifications. This is due to the requirements from most of the games. Most games in the design should include multiplayer games as assessed in the survey.	Board games are majorly played by families or groups of friends and hence the target audience is decided. Furthermore, the age restriction is simply due to the presence of small game pieces that can be hazardous to children. And finally, because the product is supposed to bring people together, all the games should be multiplayer games.
Environment	This product will have an impact on the environment as this product will be made using wood. But materials such as plastic that have a major long term impact should not be included. Environment friendly paints should be used.	The environment is extremely important for all of us and hence it is essential that we try and save the environment in every step of the way. And hence, this product should be eco-friendly.
Size	The product should be the average size of a board game. According to the research conducted the average board game is 15 x 15 inches. And hence the board game should be at least 15 x 15 inches.	The average size of a square board game ranges between 14-16 inches and hence I think that the product should be 15x15 inches so that the game does not change in any way and the game elements are not compromised.
Safety	This product should be safe for children as this product is supposed to be a family friendly game. Keeping this in mind, the edges should be smoothened.	This product also caters to children and hence safety plays a huge role in this product. And hence factors such as the edges and other sharp corners that this product may include.
Function	The function of this board is to connect people over several board games. The product should be able to include several games in an innovative and convenient manner. This game should include multiple board games in one product. Each of these games should be multiplayer games. The product should include sufficient storage space for the included game pieces.	Based on the problem that needs to be solved, the function has been determined. For this product, the purpose is to connect people and save storage. This will further mean that the games have to be multiplayer games and there have to be more than 5 games in one product along with sufficient storage.
Material	All the material used in this product should be eco friendly. It will be made out of wood for durability. The use of plastic in this product should be minimal. And finally, the entire product should be sturdy.	Appropriate materials such as design equipment should be used. This clearly shows knowledge with regards to the instruments used to make the product. Additionally, the material should be eco-friendly as the environment is extremely important for everyone.

The following table will help grade the final product based on its criteria. This table includes justification for the grade that may be given in the following Criteria.

	1-2 (Limited)	3-4 (Adequate)	5-6(Substantial)	7-8 (Rigorous)
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A E S T H E T I C S	The board includes less than 3 board games.	The board includes 3 board games.	The board includes 4 board games.	The board includes 5 or more board games.
	The board does not follow the colour scheme of beige.	The board somewhat includes a shade of beige/brown in the borders	This board is mostly a shade of beige/brown in the borders.	This board is only a shade of beige/brown to show the shade of the natural wood in the borders.
	The product includes no additional colours. (board/game pieces).	The product includes a few colours. (board/game pieces).	The product includes several colours. (board/game pieces).	Additionally, the board includes several various colours in most of the games (board/game pieces).
C O S T	The cost of the product is more than \$25	The cost of the product exceeds \$20	The cost of the product exceeds \$16	The cost of the product is less than or exactly \$16
C U S T O M E R	The game is not designed for the target audience. And is not ideal for anyone to play.	The game is designed for children over the age of 6 years as children are a part of the target audience. Some should be able to play this game.	The game is designed for children over the age of 6 years as children are a part of the target audience. But most people should be able to play this game.	The game is designed for children over the age of 6 years as children are a part of the target audience. But everyone should be able to play this game.
	This includes only 2 games.	This includes less than 3 games.	This includes less than 4 games.	This includes 5 or more games.
	None of the games are multiplayer games.	Some games are multiplayer games.	Most games are multiplayer games.	All games are multiplayer games.
E N V I R O N M E N T	This product will have a major impact on the environment.		This product will have a minor impact on the environment as this product will be made using wood.	
	Materials such as plastic that have a major long term impact are included		Materials such as plastic that have a major long term impact are not included.	
	Environmentally friendly paints such as acrylic paints were not used. Instead oil paints were used.		Environmentally friendly paints such as acrylic paints are used while painting and giving the product an aesthetic look.	
S I Z E	The length and width of the board is less than 12 inches.	The length and width of the board is more than 12 inches but less than 13 inches.	The length and width of the board is more than 13 inches but less than 14 inches.	Each board is 15 x 15 inches or more.
S A F E T Y	The product is not safe for children and does not satisfy the target audience which includes children.		This product is safe for children as this product is supposed to be a family friendly game.	
	And hence the edges in this product are not smoothened and there are rough surfaces.		Keeping this in mind, the edges are smoothened and the surfaces are not rough.	

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FUNCTION	This product includes multiple board games in one product. The product includes at least 2 games in an innovative and convenient manner.	This product includes multiple board games in one product. The product includes at least 3 games in an innovative and convenient manner.	This product includes multiple board games in one product. The product includes at least 4 games in an innovative and convenient manner.	This product includes multiple board games in one product. The product includes at least 5 games in an innovative and convenient manner.
	None of these games are multiplayer games.	Some of these games are multiplayer games.	Most of these games are multiplayer games.	Each of these games are multiplayer games.
	The product includes no storage spaces .	The product includes some storage spaces for the included game pieces.	The product includes sufficient storage spaces for the included game pieces.	The product includes many storage spaces for the included game pieces.
MATERIAL	The required wood was not utilised.	Based on the final size, the required wood slab will be required.	Based on the final size, the sufficient wood slab will be required.	Based on the final size, plenty of wood slabs will be required.
	No acrylic paints or paint brushes were used.	Acrylic Paints along with a few paints paint brushes of different sizes (2-3 round/flat brushes) will be required so that it is clearly visible.	Acrylic Paints along with the required paint brushes of different sizes (2-3 round and flat brushes) will be required so that it is clearly visible.	Acrylic Paints along with several paint brushes of different sizes (flat and round, sizes-0,2,4,6,8) will be required so that it is clearly visible.
	No Appropriate materials such as nails, hammer, and electric saws were used.	Some Appropriate materials such as nails, hammer, and electric saws will be required.	Sufficient Appropriate materials such as nails, hammer, and electric saws will be required.	More than sufficient Appropriate materials such as nails, hammer, and electric saws will be required.

B2: Design ideas:

After **creating** and **finalising** my success criteria, I **designed** 3 different designs/ideas based on several different aspects such as aesthetics, the functionality of the product and the number of games included in the product. While **designing** this product, I focused on displaying the different angles through which this game can be seen and the various setups for certain designs. I have also included a list of games in the product.

DESIGN 1

Design 1 is a foldable game. Each of these sides can overlap and be placed based on the game that the customer would like to play. This design can be flipped to play monopoly or the other game placed on the other side. As shown in the final setup, the product will look like a book. The top and bottom will have the games and the side will show the games stacked.

There are 5 games included in this design. The games included in this design are Snakes and Ladders, Scrabble, Monopoly, Chess/checkers and Ludo. All the mentioned games are multiplayer games. Snakes and Ladders can have multiple players and there is no limit but the minimum number of players required to play this game is 2 players. Scrabble, Monopoly and Ludo have a minimum requirement of 2 players and the

